

DSA Day 24: Threaded Binary Trees

Book: Data Structures Through C in Depth - Ch6 §6.13 · pp 214-224 | Code in Python

Overview

Threads replace NULL links with pointers to inorder predecessor/successor, allowing stackless traversal.

Key concepts to master

- Threads instead of NULL
- In-threaded tree
- Successor/predecessor
- Stackless inorder

Python implementation

```
# conceptually: if right thread, go to successor;  
# else go to leftmost of right subtree
```

Complexity

Inorder traversal $O(n)$ with $O(1)$ extra space.

Common pitfalls

- Mixing up thread flags with real child links

Practice

- Traverse an in-threaded tree without a stack
- What problem do threads solve?

